

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and delivered **Big Data integration pipelines** using **Apache Spark (PySpark)**, **AWS Glue**, **AWS Lambda**, and **Apache Airflow** to extract, transform, and distribute data across disparate enterprise systems from multiple file-storage formats (**Parquet**, **ORC**, **Avro**) and table formats (**Delta Lake**), reducing data integration latency by **40%** and enabling real-time operational intelligence across banking and ERP domains.
- Built scalable **data warehousing solutions** on **AWS** leveraging **EMR**, **S3**, **RDS**, and **Athena** for complex data modeling and querying; implemented **Infrastructure as Code (IaC)** using **CloudFormation** and **Terraform** to standardize deployment across geographically distributed teams, reducing infrastructure provisioning time by **35%**.
- Engineered **CI/CD pipelines** using **Jenkins**, **Git**, **Maven**, **Artifactory**, and **YAML** configuration for cloud deployments; automated build, test, and artifact management workflows supporting rapid iteration and **zero-downtime deployments** across AWS compute and data platforms.
- Developed **data security frameworks** and **data lake governance** models compliant with financial-services regulations (GDPR, audit trails, change-control management); managed complex **data integration projects** interfacing with Product Management, Integration Engineering, Quality Engineering, and System Admin teams to ensure **high-performance distributed systems** met both functional and non-functional requirements.
- Implemented **Java open-source APIs**, **REST API integrations**, and database solutions using **Oracle**, **MySQL**, **PostgreSQL**, and **DynamoDB NoSQL data stores** to connect heterogeneous enterprise systems; applied **problem-solving and technical troubleshooting** expertise to resolve complex data flow issues and optimize system interoperability, achieving **99.5% SLA compliance**.
- Leveraged **AI-assisted developer productivity tools** to accelerate development and testing cycles; demonstrated strong **analytical skills and communication abilities** in translating complex **data warehousing**, **data modeling**, and **Big Data querying concepts** to cross-functional stakeholders including banking operations teams and business intelligence consumers, enabling adoption across **12+ interconnected business systems**.

SKILLS

Big Data & Data Integration – Apache Spark (PySpark), AWS Glue, Apache Airflow, AWS EMR, Parquet, ORC, Avro, Delta Lake, Iceberg, data pipeline orchestration.

AWS Data Warehousing & Cloud Platforms – Amazon S3, Amazon RDS, Amazon Athena, AWS Lambda, AWS CloudWatch, Amazon DynamoDB, Infrastructure as Code (Terraform, CloudFormation).

Data Engineering & SQL – PostgreSQL, MySQL, Oracle, data modeling, data lake design, data security, ETL/ELT processes, complex query optimization, schema design.

CI/CD & DevOps – Jenkins, Git, Maven, Artifactory, YAML, cloud deployments, automated testing, artifact management.

Programming & API Development – Java (Spring Boot), Python (pandas, NumPy, scikit-learn), REST APIs, Scala, PySpark, open-source API standards.

Data Analytics & Business Intelligence – Power BI (DAX, Power Query), Tableau, data modeling, statistical analysis, machine learning, business intelligence integration design.

Domain & Regulatory Expertise – Financial services (SWIFT MT/MX, ISO 20022, SEPA, cross-border payments, PSD2), GDPR compliance, audit trails, ERP integration, change-control management, distributed systems architecture.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin	02/2026
BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India	01/2021

PROJECTS

Enterprise Data Warehouse & Lake Integration Platform (HCL Technologies)

- Architected and deployed a **petabyte-scale data warehouse** combining **AWS EMR, S3, RDS, Athena, and AWS Glue** to consolidate financial transaction data, customer master records, and operational metrics from **15+ disparate banking and ERP systems** into a unified **data lake** using **Delta Lake table format** for ACID compliance and schema evolution.
- Designed **PySpark-based data pipelines** running on **EMR clusters** orchestrated by **Apache Airflow** to process **500GB+ daily ingestion** across multiple file formats (**Parquet, ORC, Avro**); implemented **Infrastructure as Code (CloudFormation)** for auto-scaling cluster provisioning, reducing infrastructure costs by **28%** while maintaining **sub-2-hour SLA** for end-to-end ETL execution.
- Built **CI/CD automation** using **Jenkins, Git, Maven, and Artifactory** with **YAML-based deployment pipelines** to version-control data transformation logic and enable **canary deployments** across staging and production **AWS** environments, achieving **zero production incidents** over **18 months**.
- Implemented **data security and governance frameworks** including encryption at rest, **GDPR-compliant audit trails**, role-based access control, and data lineage tracking; collaborated with Quality Engineering and System Admin teams to ensure **99.99% uptime** and regulatory compliance across the platform.
- Developed **Athena SQL queries** and **RDS analytics databases** to enable real-time business intelligence; created **Power BI dashboards** exposing data lake metrics to **50+ business and operations stakeholders**, reducing manual reconciliation effort by **65%** and accelerating month-end close processes.

SWIFT Payments Operations Data Integration & Monitoring Dashboard (HCL Technologies)

- Engineered **real-time data integration** pipelines ingesting **SWIFT MT/MX message flows** and **ISO 20022 payment transaction data** from multiple banking channels into **PostgreSQL and Oracle data stores**; applied **complex data transformation logic** to normalize disparate payment formats and enrich transaction records with regulatory metadata.
- Built **operational monitoring dashboards** using **Power BI and Dynatrace** to provide visibility into payment transaction flow, settlement status, exception handling, and SLA compliance; delivered real-time alerting for **P1/P2 payment incidents**, reducing mean-time-to-resolution (MTTR) by **42%**.
- Designed **Data Lake integration** to archive historical payment records in **S3 using Parquet format** for compliance audit trails and GDPR data-retention policies; implemented role-based data access controls ensuring **PSD2 and regulatory confidentiality requirements** across cross-border and SEPA payment corridors.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)

- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)